Experiment: RF Transmission, Reflections, and Standing Waves

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1 Aim

To understand signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

2 Objectives

1. To determine the phase velocity and velocity factor of a signal in a coaxial cable from the frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable.
2. To study signal reflection from an open-ended cable and from a cable terminated with a 50 Ω load.
3. To estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched to a 50 Ω termination, indicating that its characteristic impedance is close to 50 Ω.
4. To calculate the inductance and capacitance per unit length using the measured phase velocity and the calculated characteristic impedance Z_0.
5. To measure the insertion loss and attenuation per unit length of the coaxial cable using a spectrum analyzer.

3 Apparatus Required

- RF Signal Generator
- Digital Storage Oscilloscope
- RF Spectrum Analyzer
- Coaxial RF Tee connector
- Coaxial cable under test (measured physical length $l = 92$ cm $= 0.92$ m)
- Precision matched 50 Ω RF termination load
- Coaxial interconnects and SMA cables

4 Theory

4.1 Transmission Lines and Reflections

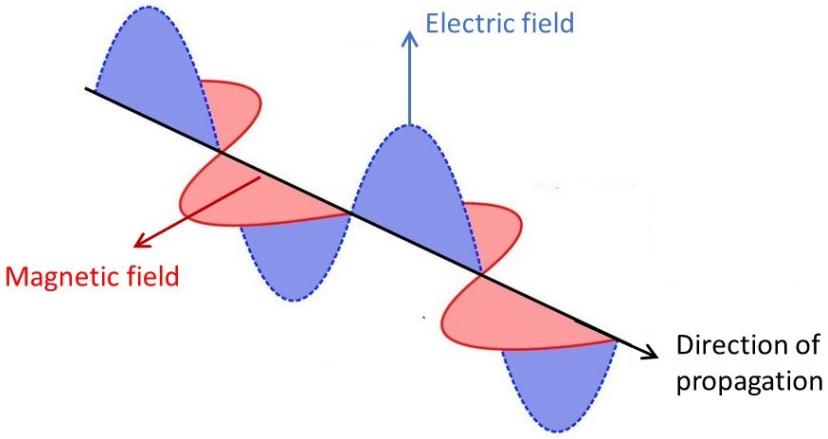


Figure 1: Transverse Electromagnetic (TEM) wave propagation with orthogonal electric ($\vec{E}$) and magnetic ($\vec{B}$) fields oriented perpendicular to the direction of propagation along the z-axis.

A transmission line is a distributed-parameter network guiding electromagnetic energy between RF systems. Coaxial lines operate in the transverse electromagnetic (TEM) mode, where both electric and magnetic field vectors lie strictly orthogonal to the longitudinal propagation axis. Due to the non-unity relative dielectric permittivity (ε_r) of the insulating spacer (typically solid polyethylene or PTFE), the phase velocity along the transmission line is lower than the free-space velocity of light ($c = 3 \times 10^8$ m/s):

$$v_p = \frac{c}{\sqrt{\varepsilon_r}} \tag{1}$$

When an incident wave strikes a load impedance Z_L terminating a line of characteristic impedance Z_0, boundary conditions dictate partial or total reflection. The terminal voltage reflection coefficient is defined as:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{2}$$

Idealized termination boundaries correspond to:

- **Open Circuit ($Z_L \to \infty$):** $\Gamma = +1$ (full in-phase reflection).
- **Short Circuit ($Z_L = 0$):** $\Gamma = -1$ (full anti-phase reflection).
- **Matched Load ($Z_L = Z_0$):** $\Gamma = 0$ (zero reflection, total transmission/absorption).

4.2 Standing-Wave Superposition at the Tee Junction

In this laboratory configuration, an oscilloscope monitors the voltage at a fixed physical location (the RF Tee connector). Because the oscilloscope channel is set to 1 MΩ, its input impedance is much larger than the 50 Ω source and cable characteristic impedance (1 MΩ $\gg$ 50 Ω). Consequently, the oscilloscope acts as an open-circuit bridge without loading the junction.

When a generator calibrated for a matched 50 Ω system delivers a nominal voltage $V_{\text{set}} = 1.0$ V, the generator's internal source emf is $E_g = 2 \times 1.0$ V $= 2.0$ V. Across a high-impedance

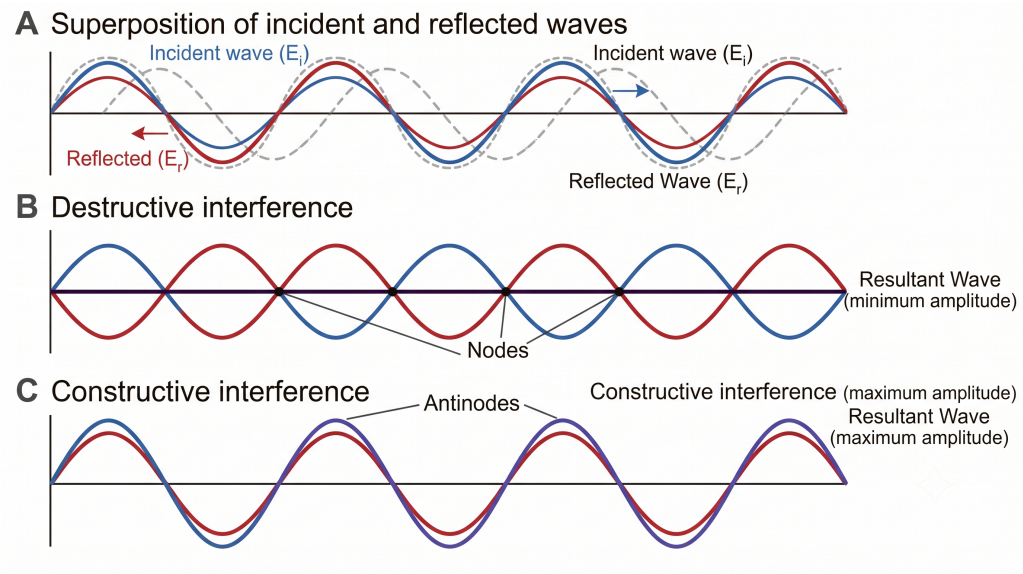


Figure 2: Standing-wave interference patterns: (A) Superposition of incident and reflected waves, (B) Destructive interference yielding minimum amplitude, and (C) Constructive interference yielding maximum amplitude.

load ($1\ \mathrm{M\Omega} \gg 50\ \Omega$), voltage doubling occurs ($V_{\mathrm{inc}} + V_{\mathrm{ref}} \approx 2.0$ V), producing a measured baseline near ≈ 1.77 V due to small internal transmission line and adapter insertion losses.

As the source frequency is varied, the round-trip electrical phase delay along the open stub alters the relative phase of the returned reflected wave:

$$V_{\mathrm{osc}}(t) = V_{\mathrm{incident}}(t) + V_{\mathrm{reflected}}(t) \tag{3}$$

The constructive and destructive envelope extremes are:

$$V_{\mathrm{max}} = |V_{\mathrm{incident}}| + |V_{\mathrm{reflected}}| \tag{4}$$

$$V_{\mathrm{min}} = |V_{\mathrm{incident}}| - |V_{\mathrm{reflected}}| \tag{5}$$

Solving for the respective wave magnitudes yields:

$$|V_{\mathrm{incident}}| = \frac{V_{\mathrm{max}} + V_{\mathrm{min}}}{2}, \qquad |V_{\mathrm{reflected}}| = \frac{V_{\mathrm{max}} - V_{\mathrm{min}}}{2} \tag{6}$$

The magnitude of the voltage reflection coefficient observed at the Tee junction is:

$$|\Gamma| = \frac{|V_{\mathrm{reflected}}|}{|V_{\mathrm{incident}}|} = \frac{V_{\mathrm{max}} - V_{\mathrm{min}}}{V_{\mathrm{max}} + V_{\mathrm{min}}} \tag{7}$$

4.3 Phase Velocity and Velocity Factor Determination

For a cable of physical length l, the round-trip delay is:

$$\tau = \frac{2l}{v_p} \tag{8}$$

Successive interference minima occur when the round-trip phase increases by 2π:

$$2\pi\Delta f\tau = 2\pi \implies \Delta f = \frac{1}{\tau} = \frac{v_p}{2l} \tag{9}$$

Rearranging gives the propagation phase velocity:

$$v_p = 2l\Delta f \tag{10}$$

The velocity factor (VF) represents the ratio of the guided phase velocity to the speed of light in vacuum ($c \approx 3.0 \times 10^8$ m/s):

$$VF = \frac{v_p}{c} \tag{11}$$

4.4 Distributed Parameters (L' and C')

For low-loss coaxial transmission lines, the phase velocity and characteristic impedance are expressed in terms of the distributed inductance per unit length L' (H/m) and capacitance per unit length C' (F/m):

$$v_p = \frac{1}{\sqrt{L'C'}}, \qquad Z_0 = \sqrt{\frac{L'}{C'}} \tag{12}$$

Decoupling these relations yields:

$$L' = \frac{Z_0}{v_p}, \qquad C' = \frac{1}{Z_0 v_p} \tag{13}$$

4.5 Cable Insertion Loss and Attenuation per Unit Length

With a spectrum analyzer referenced to a calibrated input power $P_{\text{ref}} = 0$ dBm, the insertion loss (IL) across frequency is:

$$IL \text{ [dB]} = P_{\text{ref}} \text{ [dBm]} - P_{\text{cable}} \text{ [dBm]} \tag{14}$$

The attenuation constant per unit length (α) is:

$$\alpha = \frac{IL}{l} \text{ [dB/m]} \tag{15}$$

5 Procedure

5.1 Part A: Standing Wave Profile and Phase Velocity Measurement

1. Measure and record the physical length of the coaxial cable under test ($l = 92$ cm $= 0.92$ m).
2. Assemble the Tee-junction setup (Figure 3): connect the RF Signal Generator to Port 1, the Digital Storage Oscilloscope (configured to 1 MΩ input coupling) to Port 2, and the coaxial cable under test to Port 3.
3. Leave the distal end of the test cable open-circuited.
4. Set the signal generator to output a continuous-wave (CW) sinusoidal tone at $V_{\text{in}} = 1.0$ V.
5. Sweep the source frequency from 10 MHz to 220 MHz and record peak-to-peak/envelope voltages at uniform frequency steps.
6. Identify the local constructive interference maxima ($V_{\max}$) and destructive interference minima ($V_{\min}$).
7. Determine the frequency spacing Δf between consecutive minima, and compute $v_p = 2l\Delta f$ and $VF = v_p/c$.

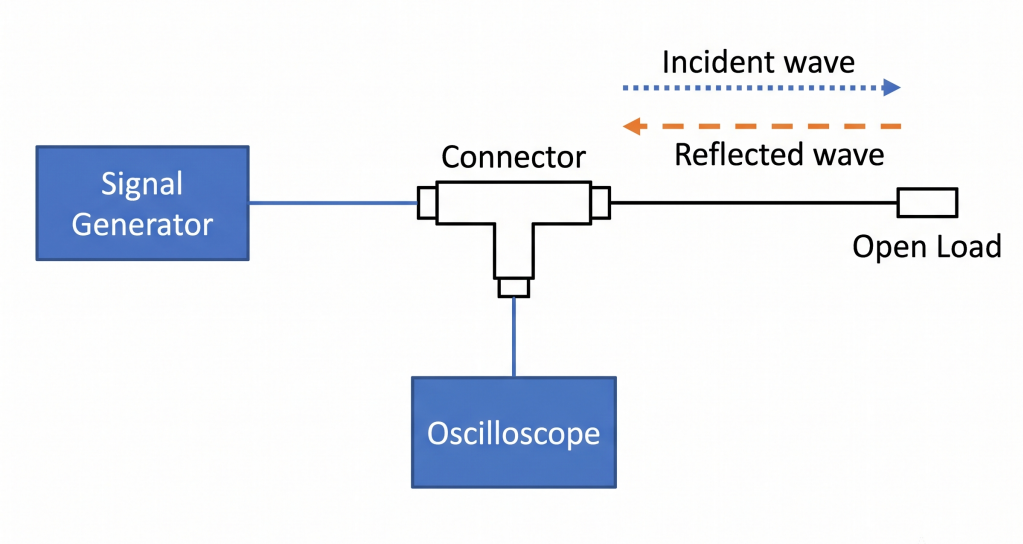


Figure 3: Experimental setup for reflection and standing-wave measurements using an RF Tee junction.

5.2 Part B: Reflection Coefficient and Characteristic Impedance Determination

1. Using the measured $V_{\max}$ and $V_{\min}$ from the open-circuit condition, calculate $|\Gamma_{\text{open}}|$.
2. Terminate the far end of the coaxial cable with a 50 Ω dummy load.
3. Sweep the generator across the same frequency band and record the new extreme voltages $V_{\max,50}$ and $V_{\min,50}$.
4. Compute the terminated reflection coefficient $|\Gamma_{50}|$ and determine the effective characteristic impedance Z_0 of the line using:

$$|\Gamma_{50}| = \left| \frac{R_L - Z_0}{R_L + Z_0} \right| \tag{16}$$

5.3 Part C: Extraction of Distributed Line Parameters (L' and C')

1. Substitute the experimental phase velocity v_p and the measured characteristic impedance Z_0 into the transmission line equations.
2. Calculate the distributed inductance per unit length $L' = Z_0/v_p$ in nH/m.
3. Calculate the distributed capacitance per unit length $C' = 1/(Z_0 v_p)$ in pF/m.

5.4 Part D: Cable Insertion Loss and Attenuation Measurement

1. Connect the RF signal generator directly to the input of the spectrum analyzer through the cable under test (Figure 4).
2. Set the reference input power on the RF signal generator to $P_{\text{ref}} = 0.00$ dBm.
3. Measure the received power P_{cable} on the spectrum analyzer at spot frequencies (50 MHz, 100 MHz, 150 MHz, and 200 MHz).
4. Calculate the insertion loss $IL = P_{\text{ref}} - P_{\text{cable}}$ and the attenuation per unit length $\alpha = IL/l$.

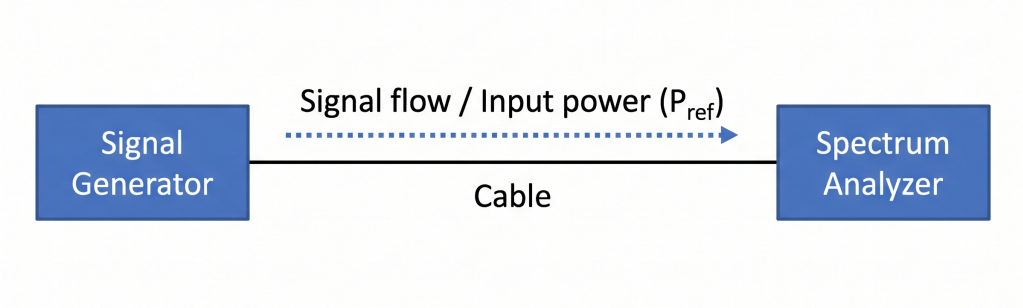


Figure 4: Experimental setup for cable insertion loss and attenuation measurement using a spectrum analyzer.

6 Observations

6.1 Physical Parameters and Baseline Measurements

- Physical length of the coaxial cable: $l = 92$ cm $= 0.92$ m
- Generator input voltage setting: $V_{\text{in}} = 1.00$ V (nominal into 50 Ω)
- Observed low-frequency baseline voltage on oscilloscope (1 MΩ mode): $V_{\text{osc}} \approx 1.77$ V
- **Reason for Baseline Mismatch:** The signal generator and coaxial cables are designed for a matched 50 Ω environment, whereas the oscilloscope has a high input impedance of 1 MΩ. Because 1 MΩ $\gg$ 50 Ω, the oscilloscope acts as an open-circuit boundary with zero phase inversion, causing the incident and reflected voltages to add in phase (1.0 V $+ 1.0$ V $= 2.0$ V theoretical). Accounting for small cable dissipation and connector losses, a net voltage of ≈ 1.77 V was observed.

6.2 Open-Ended Cable Standing-Wave Measurements

Table 1: Oscilloscope Voltage vs. Frequency for Open-Ended Cable ($l = 0.92$ m)

Frequency f (MHz)	**Observed Voltage V_{osc}**
10	1.80 V
30	1.25 V
50	1.63 V
58	178.1 mV
70	764.32 mV
90	1.17 V
110	1.62 V
112	1.64 V
130	1.37 V
150	1.28 V
170	384 mV
174	136.79 mV
220	1.19 V

- Primary cycle extremes (Cycle 1): $V_{\text{max1}} = 1.64$ V (at 112 MHz), $V_{\text{min1}} = 178.1$ mV $= 0.1781$ V (at 58 MHz)

- Secondary cycle extremes (Cycle 2): $V_{\text{max2}} = 1.19$ V (at 220 MHz), $V_{\text{min2}} = 136.79$ mV $= 0.1368$ V (at 174 MHz)
- Spacing between consecutive minima: $\Delta f = 174$ MHz $- 58$ MHz $=$ **116** MHz

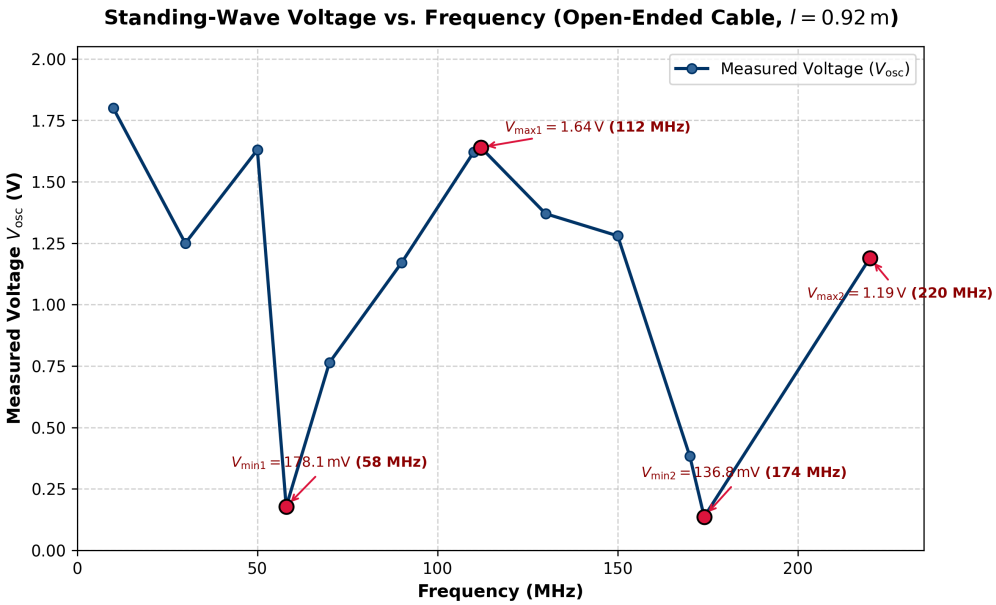


Figure 5: Frequency vs. Voltage

6.3 50 Ω Terminated Cable Standing-Wave Measurements

With a matched 50 Ω load connected at the distal end of the line:

- Observed maxima: $V_{\text{max}} = 1.44$ V (at 62 MHz), $V_{\text{max}} = 1.21$ V (at 158 MHz)
- Observed minima: $V_{\text{min}} = 890$ mV $= 0.890$ V (at 110 MHz), $V_{\text{min}} = 637$ mV $= 0.637$ V (at 226 MHz)
- Representative Cycle 1 extremes: $V_{\text{max,50}} = 1.44$ V and $V_{\text{min,50}} = 0.890$ V

6.4 Direct Cable Transmission and Attenuation Measurements

- Spectrum analyzer reference drive power: $P_{\text{ref}} = 0.00$ dBm

Table 2: Measured Cable Output Power, Insertion Loss, and Attenuation vs. Frequency

Frequency (MHz)	P_{cable} **(dBm)**	**Insertion Loss** IL **(dB)**	**Attenuation** α **(dB/m)**
50	-1.83	1.83	1.989
100	-1.82	1.82	1.978
150	-1.90	1.90	2.065
200	-2.23	2.23	2.424

7 Calculations

7.1 Phase Velocity and Velocity Factor

The frequency spacing between two successive destructive interference minima is:

$$\Delta f = 174 \text{ MHz} - 58 \text{ MHz} = 116 \text{ MHz} = 1.16 \times 10^8 \text{ Hz} \tag{17}$$

The propagation phase velocity along the coaxial line is:

$$\begin{aligned} v_p &= 2l\Delta f = 2 \times 0.92 \text{ m} \times (1.16 \times 10^8 \text{ Hz}) \\ &= 1.84 \times 1.16 \times 10^8 \text{ m/s} \approx \mathbf{2.1344 \times 10^8} \text{ m/s} \approx \mathbf{2.13 \times 10^8} \text{ m/s} \end{aligned} \tag{18}$$

The velocity factor (VF) of the coaxial cable is:

$$VF = \frac{v_p}{c} = \frac{2.1344 \times 10^8 \text{ m/s}}{3.00 \times 10^8 \text{ m/s}} \approx \mathbf{0.711} \approx \mathbf{0.71} \tag{19}$$

7.2 Voltage Reflection Coefficients

1. Open-Ended Cable Termination: Using the primary cycle extremes ($V_{\text{max1}} = 1.64$ V, $V_{\text{min1}} = 0.1781$ V):

$$|\Gamma_{\text{open}}| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{1.64 - 0.1781}{1.64 + 0.1781} = \frac{1.4619}{1.8181} \approx \mathbf{0.804} \tag{20}$$

The measured value $|\Gamma_{\text{open}}| = 0.804$ deviates slightly from the theoretical unity reflection ($|\Gamma| = 1.0$) because of finite cable dielectric and conductor attenuation over the round-trip distance.

2. 50 Ω Terminated Cable: Using the measured values with a matched load ($V_{\text{max,50}} = 1.21$ V and $V_{\text{min,50}} = 0.890$ V):

$$|\Gamma_{50}| = \frac{1.21 - 0.890}{1.21 + 0.890} = \frac{0.320}{2.100} \approx \mathbf{0.152} \tag{21}$$

The substantial reduction from $|\Gamma_{\text{open}}| = 0.804$ down to $|\Gamma_{50}| = 0.152$ demonstrates that the 50 Ω termination absorbs the majority of the incident energy.

7.3 Characteristic Impedance Determination

Using the load reflection coefficient relation with $R_L = 50$ Ω:

$$|\Gamma_{50}| = \left| \frac{R_L - Z_0}{R_L + Z_0} \right| = 0.152 \tag{22}$$

Solving for Z_0 when $Z_0 < R_L$:

$$\begin{aligned} &\frac{50 - Z_0}{50 + Z_0} = 0.152 \implies 50 - Z_0 = 0.152(50 + Z_0) \\ &50 - Z_0 = 7.6 + 0.152 Z_0 \\ &1.152 Z_0 = 42.4 \implies Z_0 = \frac{42.4}{1.152} \approx \mathbf{36.8\ \Omega} \end{aligned} \tag{23}$$

7.4 Distributed Cable Inductance and Capacitance

Using $v_p = 2.1344 \times 10^8$ m/s and the experimentally derived $Z_0 = 36.8$ Ω:

$$L' = \frac{Z_0}{v_p} = \frac{36.8 \ \Omega}{2.1344 \times 10^8 \text{ m/s}} \approx 1.724 \times 10^{-7} \text{ H/m} = \mathbf{172.4} \text{ nH/m} \tag{24}$$

$$C' = \frac{1}{Z_0 v_p} = \frac{1}{36.8 \ \Omega \times 2.1344 \times 10^8 \text{ m/s}} \approx 1.273 \times 10^{-10} \text{ F/m} = \mathbf{127.3} \text{ pF/m} \tag{25}$$

7.5 Cable Insertion Loss and Attenuation per Unit Length

For cable length $l = 0.92$ m across the observed spectrum (Table 2):

- **At 50 MHz:**

$$IL = 0.00 - (-1.83) = 1.83 \text{ dB} \implies \alpha = \frac{1.83 \text{ dB}}{0.92 \text{ m}} \approx \mathbf{1.99} \text{ dB/m} \tag{26}$$

- **At 100 MHz:**

$$IL = 0.00 - (-1.82) = 1.82 \text{ dB} \implies \alpha = \frac{1.82 \text{ dB}}{0.92 \text{ m}} \approx \mathbf{1.98} \text{ dB/m} \tag{27}$$

- **At 150 MHz:**

$$IL = 0.00 - (-1.90) = 1.90 \text{ dB} \implies \alpha = \frac{1.90 \text{ dB}}{0.92 \text{ m}} \approx \mathbf{2.07} \text{ dB/m} \tag{28}$$

- **At 200 MHz:**

$$IL = 0.00 - (-2.23) = 2.23 \text{ dB} \implies \alpha = \frac{2.23 \text{ dB}}{0.92 \text{ m}} \approx \mathbf{2.42} \text{ dB/m} \tag{29}$$

8 Results

Table 3: Summary of Measured and Derived Transmission Line Parameters

Quantity	**Symbol**	**Measured / Calculated Value**
Cable physical length	l	0.92 m
Consecutive minima frequency spacing	Δf	116 MHz
Phase velocity	v_p	2.13×10^8 m/s
Velocity factor	$VF = v_p/c$	0.71
Open-circuit reflection coefficient	$\lvert\Gamma_{\text{open}}\rvert$	0.804
50 Ω load reflection coefficient	$\lvert\Gamma_{50}\rvert$	0.152
Experimental characteristic impedance	Z_0	36.8 Ω
Distributed inductance per unit length (at 36.8 Ω)	L'	172.4 nH/m
Distributed capacitance per unit length (at 36.8 Ω)	C'	127.3 pF/m
Insertion loss at 50 MHz / 200 MHz	IL	1.83 dB / 2.23 dB
Attenuation coefficient at 50 MHz / 200 MHz	α	1.99 dB/m / 2.42 dB/m

9 Precautions

1. Keep the signal generator output amplitude strictly constant throughout the entire frequency sweep.
2. Keep input power levels below safe maximum limits to prevent overdriving the spectrum analyzer input mixer.
3. Avoid bending or flexing the coaxial cable during measurements to ensure uniform dielectric geometry and characteristic impedance.
4. Ensure secure tightening of all coaxial and Tee connectors.

10 Conclusion

1. Periodic amplitude variations on the oscilloscope confirmed the standing-wave interference resulting from the superposition of incident and reflected waves in the open-ended cable.

2. The frequency spacing of $\Delta f = 116$ MHz between consecutive destructive minima yielded a signal phase velocity of $v_p = 2.13 \times 10^8$ m/s and a velocity factor of $VF = 0.71$.

3. Terminating the cable with a 50 Ω load caused the reflection coefficient to drop significantly from $|\Gamma_{\text{open}}| = 0.804$ to $|\Gamma_{50}| = 0.152$, yielding an experimental characteristic impedance of $Z_0 = 36.8\ \Omega$.

4. The distributed line parameters were determined to be $L' = 172.4$ nH/m and $C' = 127.3$ pF/m.

5. Direct spectrum analyzer measurements showed that cable attenuation increases monotonically with frequency, rising from 1.99 dB/m at 50 MHz to 2.42 dB/m.